

Executive Report — Web3Geeks Client Inquiry Desk

Week 2 Day 5 Capstone · LangGraph + Groq + FastAPI + SQLite

1. Business goal

Inbound client messages about price, stock, comparisons, quotes, and complaints are slow to answer by hand and easy to get wrong on numbers. This system sorts the message, looks up a local product catalog, drafts a short reply from those facts, logs the ticket, and waits for human approval before sending a quote or escalating.

2. Architecture

FastAPI **/inquire** and **/approve** wrap a LangGraph pipeline: validate → classify → retrieve (CSV + calculator) → draft → quality loop → human checkpoint (quote/escalate) → apply action → finalize. Failures route to a graceful error node. Tickets land in SQLite; JSONL logs capture latency, tokens, tool calls, and errors. Full diagram: docs/architecture.md.

3. Framework choice

LangGraph was chosen over CrewAI for this desk. The problem needs explicit control, retries, and an approval interrupt — not a free-form multi-agent chat. CrewAI helped with roles in Day 4, but hierarchical runs cost more time/tokens without a clear quality win on similar catalog work. A raw ReAct loop is fine for demos, but weaker for checkpointed approval and API resume. Catalog tools and critique ideas still reuse Days 1–4.

4. Evaluation results

Six criteria (0/1): task success, factual accuracy, latency, cost, tone/quality, safety. Ten test cases include empty input, prompt-injection style text, unknown products, and a simulated catalog timeout. Latest run: **10/10** task success, mean score **6.0/6**, happy-path latency about 1.5–2.5s. Full table: **results/evaluation_results.md**.

Most common failure: catalog name mismatch (client wording ≠ CSV titles).

What to change next: fuzzy/alias matching plus top-3 catalog suggestions on miss.

5. Known limitations

- Small static CSV catalog (not live inventory).
- Quote send / escalate are placeholders (logged only, not real email/ITSM).
- MemorySaver checkpointer — open approvals do not survive process restart.
- Mostly keyword intent routing; odd phrasing can mis-route.
- Heuristic quality score, not a full judge panel.

6. Recommended next steps

- Durable checkpointer (SQLite/Postgres) for HITL across deploys.
- Stronger guardrails + PII scrubbing in logs.
- Fuzzy retrieval; sync catalog from real stock DB.
- Simple human approval UI for the quote/escalate queue.
- Grow eval set; weekly regression on model changes; follow monitoring checklist.
- Two-week pilot in shadow mode with approval required on quotes.

Run: `pip install -r requirements.txt · python evaluate.py · uvicorn api:app --port 8000`